Data Management and Sharing Plan

NSF CISE CORE (IIS) — CopernicusAI Knowledge Engine

Principal Investigator: Gary Welz | **Project Period:** 36 months | **Page Limit:** 2 pages maximum

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1. Data Types Generated

Research Data: Structured metadata objects (JSON for papers, videos, briefings, diagrams, audio); cross-modal linking data; evaluation metrics (user study data, usability metrics); collaborative validation data; system performance data (grounding rates, citation integrity); algorithmic evaluation data.

Software/Code: Metadata schema definitions, cross-modal linking algorithms, reliability mechanisms (multi-checker workflows, citation verification), evaluation frameworks, API integrations. All under version control (Git/GitHub).

Documentation: Technical documentation (architecture, API, deployment); evaluation reports; user guides; research publications and presentations.

User Study Data: Anonymized usage analytics, survey/interview data, task-based evaluation data, collaborative validation patterns. All anonymized before analysis; no PII stored without consent.

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2. Data Standards and Formats

Metadata: JSON Schema for structured objects; DOI/arXiv IDs for papers; Mermaid for process diagrams; standard transcript and audio formats.

Code: Git/GitHub; inline documentation and README; unit and integration tests; MIT License for open-source components.

Data Formats: JSON (metadata), CSV (evaluation metrics); Python/TypeScript/JavaScript (code); Markdown/PDF (documentation).

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3. Data Storage and Preservation

Storage: Google Cloud Storage (GCS) for structured data, media, and datasets; Firestore for metadata and relationships; GitHub for code. Automated backups to GCS with versioning.

Preservation: Deliverables preserved in GCS; code in GitHub with public access for open-source components; documentation in repositories and separate systems; anonymized evaluation and user study data preserved for future research.

Retention: Research data retained at least 3 years post-project (NSF requirement). Code maintained in active repositories. User study data: anonymized data retained 3 years post-project; raw data destroyed after anonymization.

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4. Data Sharing and Access

Open Source (MIT License): Metadata schemas, cross-modal linking algorithms, process visualization tools, evaluation frameworks, API specifications. Public access via GitHub (https://github.com/garywelz/copernicusai-research) and Hugging Face Spaces.

Publicly Accessible: Enhanced CopernicusAI platform, technical documentation, user guides, anonymized evaluation datasets. Publications via arXiv and institutional repositories.

Timeline: Incremental sharing during project; core open-source release by Year 2; full documentation and evaluation reports within 6 months of project completion; ongoing repository updates.

Controlled Access: Production systems and APIs with authentication and rate limiting. Anonymized user study data available upon request for research purposes.

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5. Privacy and Security

Privacy: All user data anonymized before analysis; no PII without explicit consent; informed consent for user studies; compliance with GDPR (international) and FERPA (educational data where applicable).

Security: Role-based access control; encryption at rest and in transit (HTTPS/TLS); secure authentication; continuous monitoring; standard anonymization for user study data.

IRB: All human subjects research compliant with IRB requirements; secure storage and transmission; access limited to authorized research team.

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6. Responsibilities and Compliance

PI (Gary Welz): Overall responsibility for data management and sharing; compliance with NSF policy; data quality and integrity; coordination of sharing and preservation.

NSF Compliance: Practices align with NSF data management and sharing policy. Annual and final reports will document data management activities, data generated and shared, and publication of results. Resources (GCP budgeted ~$78K over 3 years, GitHub, documentation systems) are adequate for these requirements.

Prepared by: Gary Welz, PI | **Date:** December 2025